VARUN REDDY KOKKANTI

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PROFESSIONAL SUMMARY

Results-driven AI Engineer with 5 years of experience in developing and deploying advanced AI solutions, including Generative AI and NLP pipelines. Proven ability to streamline processes and enhance decision accuracy in high-stakes environments, such as healthcare and finance. Demonstrated expertise in optimizing end-to-end workflows using LLMs, deep learning frameworks, and vector databases. Committed to fostering collaboration and delivering scalable, production-ready models for impactful outcomes.

SKILLS

- **Programming Languages**: Python, SQL
- **AI & Machine Learning**: Generative AI, NLP, Large Language Models (LLMs), Deep Learning, CNN, RNN, LSTM, Prompt Engineering
- **Frameworks & Libraries**: TensorFlow, PyTorch, Hugging Face Transformers, LangChain, LlamaIndex, FastAPI, Django, Streamlit
- **Databases**: SQL Server, MongoDB, Neo4j, Qdrant, ChromaDB
- **Data Visualization**: Tableau, Power BI, Streamlit, Plotly
- **MLOps & ModelOps**: MLflow, Kubeflow, Databricks, CI/CD, Model Versioning, Experiment Tracking
- **Cloud**: AWS (Lambda, S3), Azure (Azure OpenAI, Functions, Blob Storage)
- **DevOps & Tools**: GitHub Actions, Docker, Git, JIRA, Postman

WORK HISTORY

CDW — **Jul 2023 - Present**
GENERATIVE AI ENGINEER — *Irving, TX*

- Developed and deployed production-ready Generative AI solutions using LLMs, LangChain, and Azure OpenAI, streamlining healthcare claims processing and enhancing decision accuracy.
- Engineered Retrieval-Augmented Generation (RAG) pipelines with Neo4j and Qdrant, improving semantic search performance and response relevance in high-stakes environments.
- Constructed scalable NLP pipelines for extracting information from unstructured claims data, increasing automation and reducing manual processing by 40%.
- Designed and integrated secure, compliant AI systems utilizing FastAPI, Azure Functions, and HIPAA-aligned validation tools like Guardrails AI.
- Directed model evaluation and tuning efforts using metrics such as ROUGE, BERTScore, and hallucination tracking via LangSmith, boosting response reliability and model trust.
- Collaborated with cross-functional teams to manage end-to-end deployment, observability, and continuous improvement of LLM workflows in production.
- Implemented real-time document ingestion pipelines with Azure Event Grid and Blob Storage, enabling dynamic vector index updates and improved response freshness.
- Optimized LLM performance through advanced prompt engineering, temperature tuning, and model configuration to enhance contextual accuracy and reduce hallucination rates.

SoftData Inc — **Jul 2022 - May 2023**
GENERATIVE AI ENGINEER — *St. louis, MO*

- Contributed to the development and deployment of an SVM-based fraud detection system, improving detection accuracy and reducing manual review workload.
- Engineered key feature extraction from transactional and behavioral data, enhancing model performance and robustness.
- Supported hyperparameter tuning and model validation using grid search and cross-validation, boosting fraud detection efficiency by 20%.
- Built automated data preprocessing pipelines to ensure high-quality, real-time inputs for model training.
- Collaborated with compliance and business teams to ensure model explainability and adherence to regulatory standards.
- Produced clear documentation and dashboards to communicate fraud trends and model insights to non-technical stakeholders.
- Learned ML best practices, model validation, and Agile workflows under the mentorship of senior data architects, fostering team collaboration and knowledge sharing.

Creative Shree Vikas — **Jan 2020 - Dec 2021**
Machine Learning Engineer — *Bangalore, India*

- Collaborated on the Customer Segmentation and Credit Utilization Optimization project at MONETA Money Bank, aimed at enhancing customer insights and optimizing credit product strategies.
- Applied K-Means clustering to segment customers based on behavioral and demographic attributes, delivering clear and actionable group profiles for targeted marketing and risk management.
- Engineered key features including account age, average utilization rate, transaction frequency, overdue patterns, and payment-to-balance ratios to improve model accuracy.

- Assessed supervised learning models such as Decision Trees, Logistic Regression, and Gradient Boosting to predict credit overutilization risk and repayment behaviors.
- Implemented Gradient Boosting Classifier in production for credit utilization prediction, chosen for its superior recall and AUC metrics.
- Provided business stakeholders with insights derived from cluster analysis, contributing to a 12% uplift in personalized campaign engagement.
- Developed reproducible Python workflows for model training, evaluation, and integration of A/B testing using scikit-learn and pandas.
- Supported preparation of technical documentation and dashboards for internal quality assurance and executive leadership reviews.

EDUCATION

University of Texas At Arlington, Arlington, TX **Jan 2022 - May 2023**
Master of Science, Data Science

REVA University, Bangalore, India **May 2017 - Apr 2021**
Bachelor of Technology, Computer Science

CERTIFICATIONS & AWARDS

- Publication: "OSEMOSYS Modeling Primer in Gurobipy” – ISCT Journal
- **Codecademy – React**
- **TCS - Data Visualization**
- **Cisco Networking Academy**